

GeHirNet: A Sex-Aware Hierarchical Model for Fair and Accurate Voice Pathology Classification

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Abstract—AI-based voice analysis shows promise as a non-invasive tool for disease diagnostics, yet existing systems often fail to generalize due to sex-related acoustic differences and scarce data for rare pathologies. We propose a clinically motivated two-stage framework that first captures sex-specific vocal representations using ResNet-50 on Mel spectrograms, followed by sex-conditioned disease classification. We address class imbalance through multi-scale resampling and time-warping augmentation. Evaluated on a merged dataset from four public repositories, our framework achieves state-of-the-art performance, attaining 97.2% accuracy and 94.5% MCC on average, consistently outperforming a single-stage baseline with an average MCC improvement of approximately 3%. Fairness evaluation shows reduced sex-based performance gaps, while feature representations reveal sex-specific patterns that provide clinical evidence supporting a hierarchical, sex-aware design. Together, these results demonstrate that sex-aware hierarchical modeling improves both classification accuracy and fairness in voice pathology detection, with potential applications in telemedicine and clinical decision support.

Index Terms—Voice pathology, Sex-aware learning, Hierarchical models, Deep learning, Audio biomarkers, Representation learning, Fair AI

I. INTRODUCTION

Voice pathologies, caused by factors such as infections, vocal fatigue, or neurological conditions such as muscular dystrophy, disrupt normal vocal fold vibration, resulting in strained, weak, or hoarse voices that degrade voice quality [1]. Traditional diagnosis relies on invasive clinical procedures like laryngoscopy and stroboscopy, which require specialized equipment, cause patient discomfort, and incur high costs [2].

Recent advances in signal processing and AI provide a promising non-invasive alternative: voice-based detection. By analyzing acoustic features from speech recordings, machine learning models have shown promise in distinguishing normal voices from pathological ones [3]. Recent studies reported 98% accuracy with the MEEI database [4] and 95.41% accuracy with the SVD dataset [5].

While current approaches perform well in broad classifications (e.g., pathological vs. healthy), their accuracy declines when discriminating between specific pathology subtypes. One early study reported accuracies of only 67.8% for male patients and 52.5% for female patients when identifying five distinct laryngeal pathologies [6]. The diversity of diseases complicates voice pathology classification. Another study achieved

95% accuracy in detecting laryngeal disorders but only 85% for Parkinson’s disease [7], further underscoring the difficulty in distinguishing between specific conditions.

These shortcomings are worsened by challenges such as biological sex disparities and data imbalance. Sex significantly influences voice characteristics, as male and female voices differ in pitch and frequency distribution [8]. Recent studies have explored sex-related patterns with deep learning models, such as the convolutional neural network (CNN) in applications such as emotion detection [9]. Narrowing the search space to a particular sex has enhanced classification accuracy [10]. These advancements provide a novel perspective for leveraging sex-specific features in voice pathology classification. Researchers developed a cascaded model incorporating sex classification as a preliminary step, achieving 88.38% accuracy for binary pathology detection [11], demonstrating the benefits of sex-aware modeling.

Another critical challenge is the class imbalance commonly present in voice pathology datasets, which is often overlooked. High overall accuracy in such imbalanced datasets may not reflect true robustness, as models tend to be biased toward the majority class (e.g., healthy controls) and underperform on rare diseases. This issue has received increasing attention, with various methods proposed to mitigate data imbalance [12], highlighting the need for approaches that ensure reliable performance across all classes.

A recent review highlights the need for vowel and sex separation, while addressing the uneven distribution of rare pathologies [13]. Accordingly, we hypothesize that the class imbalance commonly found in voice pathology datasets, often overlooked by researchers, is further aggravated by sex disparities. Differences in voice characteristics between males and females can result in unequal representation and introduce performance biases across various pathologies.

To address these challenges, we propose a novel hierarchical framework **GeHirNet** that first distinguishes male and female pathologies from healthy controls, then classifies specific diseases separately for each sex. To the best of our knowledge, this is the first framework to integrate sex-based differentiation in the initial stage of multi-class voice pathology classification. Our framework analyzes sustained vowel /a/ recordings from multiple datasets, including Coswara, SVD, ALS, and PC-GITA, to detect a diverse range of pathologies, including

COVID-19, Parkinson’s Disease, Vocal Cord Paresis, Dysphonia, Laryngitis, and Amyotrophic Lateral Sclerosis (ALS).

By explicitly modeling sex-specific vocal characteristics, GeHirNet improves classification performance and supports fairness, reducing sex-based performance gaps. Analysis of the learned feature representations using similarity scores and Mel spectrogram statistics shows that early layers reflect shared vowel characteristics, while deeper layers reveal sex-specific patterns corresponding to anatomical and acoustic differences, supporting the sex-aware hierarchical design.

The key contributions of this paper are as follows:

- **Hierarchical sex-aware architecture:** Two-stage framework integrating sex-specific patterns in stage one, followed by multi-class pathology classification.
- **Class imbalance handling:** Multi-scale resampling and novel time-warping augmentation improve robustness for underrepresented classes.
- **Comprehensive evaluation:** Validated on four public datasets across multiple pathologies, achieving state-of-the-art performance.
- **Representation-based insight:** CKA and Mel spectrogram analysis show deeper layers capture sex-specific, clinically meaningful patterns.
- **Fairness evaluation:** Incorporating a sex-aware stage reduces sex-based performance disparities, supporting equitable diagnostics.

The code is available at GeHirNet’s GitHub.

II. METHODS

A. Dataset

We used four publicly available datasets. This study only considered vowel /a/ recordings from these datasets.

Coswara (English) [14]: Designed for COVID-19 classification, it includes vowel /a/ recordings from 341 COVID-19 patients (212 male, 129 female) and 924 healthy individuals (701 male, 223 female).

ALS (Russian) [15]: Designed for ALS diagnosis, a neurodegenerative disorder, it includes vowel /a/ recordings from 39 healthy individuals (22 male, 17 female) and 15 ALS patients (7 male, 8 female).

PC-GITA (Spanish) [16]: Developed for Parkinson’s disease research, it includes 300 vowel /a/ recordings from 100 participants (each providing three samples), with 50 healthy controls (25 male, 25 female) and 50 Parkinson’s patients (25 male, 25 female).

SVD (German) [17]: This dataset includes over 70 voice disorders. We considered the most represented pathology categories: healthy controls (259 male, 428 female), Laryngitis (50 male, 32 female), Vocal cord paresis (70 male, 127 female), and Dysphonia (99 male, 174 female). In our work, the Dysphonia category merges seven subtypes from the SVD dataset: Dysphonia, Functional Dysphonia, Spasmodic Dysphonia, Psychogenic Dysphonia, Hypofunctional Dysphonia, Hypotonic Dysphonia, and Juvenile Dysphonia.

We selected these datasets to enable robust sex-specific pathology analysis: (1) voice recordings from speakers of

diverse language backgrounds to examine sex-based vocal patterns, (2) standardized sustained vowel /a/ segments to isolate fundamental voice features while controlling for linguistic variability, and (3) comprehensive pathology coverage with healthy controls to evaluate model generalizability. This minimizes population-specific biases while allowing systematic investigation of sex-dependent pathological signatures.

The distribution of original recordings in the merged dataset is summarized in Table I. Healthy control recordings come from four datasets, while pathological recordings cover six diseases: COVID (Coswara), ALS (ALS), Parkinson’s (PC-GITA), Laryngitis, Vocal Cord Paresis, and Dysphonia (SVD).

B. Audio Pre-processing

1) *Silence Removal:* Audio recordings in the datasets, particularly in Coswara, frequently contain silent periods. We followed established pre-processing practices in prior work [18] by implementing silence removal as our initial processing step. We first addressed inconsistencies by removing silent segments based on Root Mean Square (RMS) energy [19]. The audio signal was segmented into overlapping frames with a window size of $W = 2048$ with a hop size of $H = 512$. For each frame i , the RMS energy $E_{\text{RMS}}^{(i)}$ is computed as (1).

$$E_{\text{RMS}}^{(i)} = \sqrt{\frac{1}{W} \sum_{n=0}^{W-1} x_i^2[n]}, \quad (1)$$

where $x_i[n]$ represents the n -th sample in the i -th frame. Frames with RMS energy below an empirically determined threshold of 10^{-3} were classified as silent. Since our data consists of only vowel recordings, we removed silent segments and concatenated voiced segments to reconstruct the audio.

Direct concatenation, however, introduces artifacts, i.e., sharp transitions at segment boundaries. To prevent this, we applied linear crossfade, shaping fade profiles for smooth transitions and minimizing discontinuities [20]. Linear crossfade interpolates between the overlapping regions of neighboring segments. Given a crossfade interval V , the crossfaded output $y[n]$ for each sample point $n \in [0, V - 1]$ is computed as (2).

$$y[n] = (1 - \alpha_n) \cdot x_1[n] + \alpha_n \cdot x_2[n] \quad (2)$$

where $x_1[n]$ represents n -th sample from the last V samples of the previous segment, $x_2[n]$ represents the n -th sample from the first V samples of the next segment. $\alpha_n = \frac{n}{V-1}$ is the linear interpolation weight. We set the crossfade interval V as 512.

2) *Outlier Removal:* After removing silence, we generated Mel spectrograms for all recordings using a window size of 2048 and a hop size of 512 for visual inspection [21]. As we expect sustained vowel recordings, we manually excluded recordings with abnormal spectral energy concentration, transient impulse artifacts, or aperiodic waveforms.

3) *Normalization*: Min-Max normalization was applied to all valid recordings to compress the amplitude into the range [0, 1], to standardize the data across multiple datasets and to ensure consistency in amplitude levels. For an audio signal $x[n]$ of length N , the normalized signal $\tilde{x}[n]$ is computed as (3).

$$\tilde{x}[n] = \frac{x[n] - \min(x)}{\max(x) - \min(x)} \quad (3)$$

4) *Segmentation*: Finally, we segmented each audio file into 1-second clips using a sliding window with 0.6-second overlap (e.g., [0,1]s, [0.4,1.4]s, ...) to capture temporal context [22].

C. Data Augmentation

From the segment distribution, we observed a significant imbalance between healthy control and specific pathological diseases. To address this, we handled the data imbalance with data augmentation techniques.

1) *Multi-scale Resampling*: Data resampling is a common technique for mitigating data imbalance. In particular, we use sample-rate conversion to generate additional instances of minority classes and equalize their representation. In our dataset, ALS and PC-GITA have a sampling rate of 44.1 kHz, SVD uses 50 kHz, and Coswara includes both 48 kHz and 44.1 kHz rates. We applied sample-rate conversion to the classes with fewer samples by selecting a specific sampling rate from a range of 40 kHz to 50 kHz, with 125 Hz intervals. This process includes both upsampling and downsampling, with a sinc filter applied as an anti-aliasing filter. Resampling was repeated until the sample count for the underrepresented classes matched that of the most frequent class.

2) *Time Warping*: Time warping was originally applied to spectrograms by swapping two blocks of the same length from the time or frequency dimensions [23]. To compare fairly with resampling, we applied time warping directly to audio, which is plausible since vowel recordings are sustained and lack semantic information. We split each 1-second vowel recording into five segments and randomly shuffled their order before reassembly. To create smooth transitions between segments while preserving pathological characteristics, we applied crossfade with a short interval $V = 32$ during time warping augmentation.

D. Mel Spectrogram

Since Mel spectrograms effectively capture spectral information and speech variations [24], we converted all audio recordings into Mel spectrograms $\{S_1, S_2, \dots, S_N\}$. $S_i(f, t)$ is the energy at frequency f and time t for the i -th sample, where f denotes the Mel filter banks (1 to 128), t denotes the time frame (1 to 98) [25]. Each time frame corresponds to a short segment of the audio, determined by the window and hop size, capturing the temporal evolution of the speech signal. For 50 kHz audio, we used a window length of 2048 and a hop length of 512. To ensure consistency across different sampling rates (44.1 kHz, 48 kHz, and resampled audio), we adjusted the short-time Fourier transform (STFT) parameters to generate

consistent Mel spectrograms from 1-sec audio segments. This approach ensures consistent input size for feature extraction and model training, with slight variation in spectral resolution. As a result, the model received identical input representations, minimizing distortion and maintaining feature alignment.

E. Backbone Model

CNNs are highly effective in capturing spatial dependencies within two-dimensional features like Mel spectrograms and have proven useful for detecting subtle differences in pathological voices [26]. Among CNN architectures, ResNet is particularly well-suited for pathological voice detection, achieving 98.13% accuracy [24]. The model ResNet-50 was pretrained on the ImageNet dataset [27], and we modified the input depth to 1 to match the single-channel Mel spectrogram input. With Mel spectrograms as input representations, all layers were trained during model training.

F. Two-Stage Hierarchical Architecture

Using ResNet as the backbone, we propose a two-stage hierarchical architecture **GeHirNet**, as illustrated in Fig. 1. The first stage focuses on sex-based *Pathology Detection (Classifier PD)*, classifying samples into four target classes: male healthy control (HC_M), female healthy control (HC_F), male pathology (P_M), and female pathology (P_F). The male and female healthy control groups are later merged into a single healthy control category.

In the second stage, we classify specific diseases separately within the male and female pathology groups. We achieve this by *Pathology Classification, Classifier FP* for female pathologies and *Classifier MP* for male pathologies. *Classifiers FP* and *MP* distinguish among six diseases (COVID-19, Parkinson's, Dysphonia, Vocal Cord Paresis, Laryngitis, ALS). The final disease classification result is obtained by merging the predictions from both sex-specific classifiers. All three classifiers, *Classifiers PD*, *MP*, and *FP*, used the same ResNet-50 backbone and were trained separately.

G. Experiment Setup

To validate the effectiveness of the two-stage architecture and data augmentation techniques, we conducted four experiments independently. We split the dataset into a training set (80%) and a test set (20%) for evaluation. The split was stratified to ensure the same class distribution across training and testing data. All experiments used the same training and testing datasets.

- **Exp 1 (Baseline)** : Single-stage
- **Exp 2 (GeHirNet)** : Two-stage
- **Exp 3.1 (GeHirNet*)** : Two-stage + Resampling
- **Exp 3.2 (GeHirNet**)** : Two-stage + Time warping

In **Exp 1** and **Exp 2**, no data augmentation was applied. After audio pre-processing, recordings were directly converted into Mel spectrograms as input images. **Exp 1** served as our **Baseline** and used a single-stage architecture to classify healthy controls and six specific pathologies, while **Exp 2** employed two-stage hierarchical framework **GeHirNet**, where

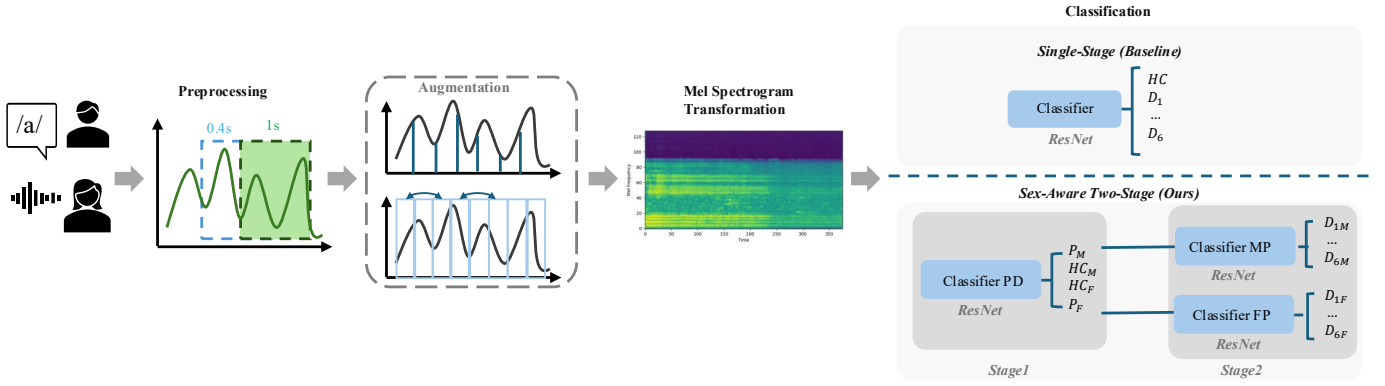


Fig. 1. Training pipeline of data processing, experiment setup, and model architecture

the sex-based *Classifier PD* preceded separate classifiers for specific diseases (*Classifiers MP and FP*). Both single-stage and two-stage architectures used ResNet-50 as the backbone. We compare **GeHirNet** with **Baseline** to evaluate the effectiveness of our two-stage hierarchical architecture.

We designed two experiments, **Exp 3.1** and **Exp 3.2**, to validate the effectiveness of data augmentation in addressing data imbalance. Data augmentation was applied only to the training set, keeping the test data unchanged. **Exp 3.1** and **Exp 3.2** incorporated resampling and time warping, respectively, for training data. After audio pre-processing, **Exp 3.1 (GeHirNet*)** applied resampling before converting audio to Mel spectrograms, while **Exp 3.2 (GeHirNet**)** applied time warping before conversion. Since *Classifiers PD, MP, and FP* were trained separately, augmentation was performed separately at two stages: *Classifier PD* was balanced to match healthy control samples, while *Classifiers FP and MP* were balanced to the most frequent disease (COVID-19).

H. Training & Evaluation

Taking Mel spectrograms as input, we trained the models using 5-fold cross-validation. We explored hyperparameter learning rates (η): $\{10^{-3}, 10^{-4}, 10^{-5}\}$, batch sizes (B): $\{32, 64\}$, training epochs (E): $\{10, 20, 30\}$. The Adam optimizer was used with default parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and no weight decay. All experiments were repeated using three different random seeds (40, 41, and 42). Reported results correspond to the mean and standard deviation across these runs. Experiments were conducted on a single NVIDIA L4 GPU with PyTorch 2.6.0 and CUDA 12.4.

The optimization minimizes cross-entropy loss, which simultaneously maximizes the likelihood of correct class predictions and penalizes incorrect classifications. The labels of training data are encoded as one-hot vector $\mathbf{y}$.

$$\mathbf{y} = [y_1, y_2, \dots, y_C]$$

The output dimension C of $\mathbf{y}$ varies by experiment:

- **Exp 1** (Single-stage): $C = 7$ for six pathological diseases and healthy controls

- **Exp 2, 3.1 & 3.2** (Two-stage):

- First stage (*Pathology Detection*): $C = 4$ for male and female healthy control (HC_M, HC_F), male and female pathology (P_M, P_F).
- Second stage (*Pathology Classification*): $C = 6$ for sex-specific pathologies, $D_{1M}, \dots, D_{6M}$ for male and $D_{1F}, \dots, D_{6F}$ for female.

The predicted probability distribution is denoted as $\hat{\mathbf{p}} = [\hat{p}_1, \hat{p}_2, \dots, \hat{p}_C]$ and the categorical cross-entropy loss for a single sample is defined as (4).

$$\mathcal{L}^{(n)} = - \sum_{i=1}^C y_i^{(n)} \log(\hat{p}_i^{(n)}) \quad (4)$$

Averaging over the batch size B , the total loss becomes:

$$\mathcal{L} = \frac{1}{B} \sum_{n=1}^B \mathcal{L}^{(n)} \quad (5)$$

The optimal hyperparameter was selected based on the average Matthews Correlation Coefficient (MCC) (6) across validation folds, where TP , FP , TN and FN are true positives, false positives, true negatives and false negatives.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (6)$$

Using the best hyperparameters (η, B, E^*), we retrained the final model in the complete training set.

Finally, we assessed the model's performance on the unseen test data. We evaluated accuracy (proportion of correct predictions), weighted F1-score (harmonic mean of precision and recall, weighted by class frequency) (7) and MCC (balanced metric considering all confusion matrix scores).

$$weightedF1 = \frac{\sum_{i=1}^N w_i \cdot \left(2 \cdot \frac{TP_i}{2TP_i + FP_i + FN_i} \right)}{\sum_{i=1}^N w_i} \quad (7)$$

This metric combination was selected as accuracy provides an easily interpretable baseline, weighted F1-score specifically

handles our dataset's class imbalance by weighting minority classes, and MCC serves as our primary robust metric since it reliably evaluates performance and remains invariant to class distribution, particularly crucial for our clinical diagnostic application with imbalanced data [28].

I. Ablation Studies

To evaluate the contribution of individual components of GeHirNet and assess the robustness of the proposed design choices, we conducted two ablation studies and comparative experiments.

1) *Baseline with sex as an input feature*: We evaluated a single-stage baseline augmented with sex information using an input-level early fusion strategy. Sex was encoded as a scalar (male = 0, female = 1), expanded into a constant time frame, and concatenated to the Mel spectrogram along the temporal axis (from 128×98 to 128×99). The resulting representation was fed into a ResNet-50 backbone, allowing joint learning of acoustic features and sex context without hierarchical branching.

2) *Increased network depth in the baseline model*: To examine whether the observed improvements were primarily attributable to increased model capacity rather than the hierarchical design, we augmented the baseline classifier with two additional fully connected hidden layers (512 and 128 neurons), while keeping all other components unchanged.

J. Feature Representation & Fairness

1) *Mel Spectrogram Analysis*: To investigate disease- and sex-specific acoustic patterns, we first analyzed Mel spectrograms at the group level. For each subgroup (e.g., sex $\times$ disease), we computed the mean Mel spectrogram by averaging the time-frequency representations across all participants in the subgroup.

$$\bar{S}(f, t) = \frac{1}{N} \sum_{i=1}^N S_i(f, t) \quad (8)$$

To examine whether sex-specific acoustic characteristics were present within each disease, we projected participant-level features into 2D space using uniform manifold approximation and projection (UMAP).

Next, for each participant, we computed the mean power (in dB) of the Mel spectrogram across all time-frequency bins. We then derived group-level statistics by calculating the mean and standard deviation of these mean power values across all participants.

$$\text{Mean power (dB)} = \frac{1}{N \times M} \sum_{f=1}^N \sum_{t=1}^M 10 \cdot \log_{10}(S(f, t)) \quad (9)$$

where N and M are numbers of frequency bins and time frames.

To compare mean power between sexes for each disease, we performed a t-test when both groups satisfied normality assumptions, and reported the 95% confidence interval (CI)

of the mean difference. When normality assumptions were not met, we applied the Mann–Whitney U test as a non-parametric alternative and estimated the 95% CI of the median difference using bootstrap resampling (10,000 iterations). This provides an initial view of sex-related acoustic differences that may influence pathology classification.

2) *Feature Representation*: While Mel spectrograms capture low-level acoustic differences, deeper neural network layers may encode more abstract, clinically relevant representations. To explore these higher-order sex differences, we analyzed the similarity between the neural representations learned by the male-specific (*Classifier MP*) and female-specific (*Classifier FP*) networks using Centered Kernel Alignment (CKA) [29]. Specifically, ResNet was grouped into the convolution stem, four residual blocks, pooling, and fully connected layers, and CKA scores were computed for each group.

Given n input samples, let $X \in \mathbb{R}^{n \times d_1}$ and $Y \in \mathbb{R}^{n \times d_2}$ denote feature matrices extracted from *Classifiers MP* and *FP* respectively. The CKA similarity measure is computed as (10).

$$\text{CKA}(X, Y) = \frac{\langle K_c, L_c \rangle_F}{\|K_c\|_F \|L_c\|_F} \quad (10)$$

where:

- $\tilde{X} = (I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)X$ (centered features)
- $K_c = H K H^\top$ with $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ (double center)
- $K = \tilde{X} \tilde{X}^\top$, $L = \tilde{Y} \tilde{Y}^\top$ (Gram matrices)
- $\langle A, B \rangle_F = \text{tr}(A^\top B)$ (Frobenius inner product)

This allowed us to quantify which layers encode shared versus sex-specific features, helping to interpret how the model leverages sex-related cues for pathology classification.

3) *Fairness Evaluation*: To assess the impact of sex-aware modeling on equitable performance, we evaluated fairness across biological sexes. We computed Equalized Odds (EOdds) for each disease, defined as the average disparity in true positive rates (TPR) and false positive rates (FPR) between male and female patients [30]. This metric quantifies whether the model's predictions are consistent across sexes, providing a direct measure of bias mitigation achieved by the two-stage hierarchical framework.

$$\text{Fairness}_c^{\text{EOdds}} = \frac{1}{2} \left(|\text{TPR}_{M,c} - \text{TPR}_{F,c}| + |\text{FPR}_{M,c} - \text{FPR}_{F,c}| \right) \quad (11)$$

where TPR and FPR for sex s and disease c are:

$$\text{TPR}_{s,c} = \frac{TP_{s,c}}{TP_{s,c} + FN_{s,c}}, \quad s \in \{M, F\} \quad (12)$$

$$\text{FPR}_{s,c} = \frac{FP_{s,c}}{FP_{s,c} + TN_{s,c}}, \quad s \in \{M, F\} \quad (13)$$

We then calculated the overall MacroTPR disparity across all diseases, which captures whether males and females have similar probabilities of being correctly classified on average. A lower MacroTPR disparity indicates more equitable classification across sexes.

TABLE I
CLASS DISTRIBUTION OF THE DATASET.

	Total	Female	Male
<i>HC</i>	1800 (1675) ^a	743 (696)	1057 (979)
Pathology	1058 (987)	545 (517)	513 (470)
COVID-19	341 (283)	129 (109)	212 (174)
Parkinson's	150 (149)	75	75 (74)
Dysphonia	273 (267)	174 (170)	99 (97)
Vocal Cord Paresis	197 (191)	127 (123)	70 (68)
Laryngitis	82	32	50
ALS	15	8	7

^a Number of original recordings and post-outlier removal recordings (in parentheses). Unparenthesized ones had no removals.

$$\text{Fairness}_{\text{MacroTPR}} = \frac{1}{7} \sum_{c=1}^7 |\text{TPR}_{M,c} - \text{TPR}_{F,c}| \quad (14)$$

We compared both EOdds and MacroTPR disparity between our two-stage sex-aware GeHirNet, its variants GeHirNet* and GeHirNet**, and the baseline one-stage model to assess whether the hierarchical design reduces sex-based disparities.

III. RESULTS

A. Data Distribution

During the preprocessing, we applied silence removal and crossfade to 121 samples, 7 from PC-GITA and 114 from Coswara. After outlier removal, the final dataset comprised 1675 healthy recordings and 987 pathology recordings. Removal rates were balanced across subgroups: 6.33% for female healthy, 5.14% for female pathology, 7.38% for male healthy, and 8.38% for male pathology. The complete distribution is summarized in Table I.

After the segmentation, the dataset contained 9250 (68.5%) audio segments for healthy control, 1877 (13.89%) for COVID-19, 1075 (7.96%) for Parkinson's, 597 (4.42%) for Dysphonia, 412 (3.05%) for Vocal Cord Paresis, 171 (1.27%) for Laryngitis, 127 (0.94%) for ALS.

B. Comparison with Baseline

Table II illustrates the evaluation metrics (mean $\pm$ standard deviation) over three runs with different random seeds (40, 41, and 42) for each experiment. The two-stage architecture (**GeHirNet**) outperformed the single-stage model (**Baseline**), yielding approximately a 1% improvement in MCC (0.9294 vs. 0.9190), accuracy (0.9647 vs. 0.9598), and F1 score (0.9644 vs. 0.9590).

After incorporating data augmentation, the two-stage model with resampling (**GeHirNet***) achieved an MCC of 0.9289, an accuracy of 0.9646, and an F1 score of 0.9641, performing comparably to **GeHirNet**. The two-stage model with time warping (**GeHirNet****) achieved the highest performance with lower variance than the baseline, obtaining an MCC of 0.9449 ± 0.0074 , an accuracy of 0.9724 ± 0.0037 , and an F1 score of 0.9723 ± 0.0038 .

Taking MCC as the primary metric, **GeHirNet**** consistently outperformed the baseline across all three random seeds,

achieving an average improvement of approximately 3% with low variance. These results highlight the effectiveness of the two-stage hierarchical architecture in improving classification performance, with time-warping-based data augmentation providing additional gains.

C. Ablation Studies

Table III summarizes the ablation results. We report results from the run with random seed 42 for the ablation studies. Adding sex as an input-level feature resulted in only marginal changes in baseline performance, suggesting that naïve conditioning is insufficient to capture sex-dependent acoustic variations. Increasing network depth led to slight performance gains; however, these improvements remained below those achieved by the proposed two-stage GeHirNet. These findings indicate that the performance gains are attributable to the hierarchical sex-aware design rather than increased model capacity alone.

D. Comparison with Prior Work

Our framework achieves better classification performance across pathologies compared to sex-agnostic models. While recent work reported 89.47% accuracy for four-class discrimination (healthy vs. cyst, polyp, and paralysis) [31], our sex-aware architecture demonstrates competitive performance, achieving an accuracy of 97.63%, F1 of 97.61% and MCC of 95.25%.

To address concerns regarding potential discrepancies in disease classification, particularly between functional, organic pathologies and neurological disorders, we evaluated the performance of our model on a disease-specific basis. Table IV compares the accuracy of our framework with prior state-of-the-art (SOTA) methods, evaluated on the same dataset. We report results from the run with random seed 42 for this comparison. We benchmark three variants: single-stage **Baseline**, two-stage architecture **GeHirNet**, and its enhanced variant with time warping, **GeHirNet****. Results demonstrate that **GeHirNet** matches or exceeds SOTA performance across all six target diseases, with **GeHirNet**** achieving the highest overall accuracy.

Beyond disease-specific evaluations, we evaluated detection accuracy of healthy controls across all six diseases. **GeHirNet** achieved an accuracy of 97.6% in detecting healthy controls, while **GeHirNet**** improved this to 99.3%. This highlights the model's ability to generalize and detect healthy controls across different datasets, rather than overfitting to specific data. It suggests that our model learns underlying pathological patterns, enabling cross-dataset generalization and mitigating issues like shortcut learning.

E. Feature Representation & Fairness

We first examined disease- and sex-specific patterns in the input data by computing group-level statistics of Mel spectrograms. As shown in Fig. 2, acoustic characteristics differ between males and females across pathologies. In UMAP projections, female and male participants form largely separate

TABLE II
EXPERIMENT RESULTS

Exp	Accuracy	F1	MCC
Exp1 (Baseline)	0.9598 $\pm$ 0.0063	0.9590 $\pm$ 0.0068	0.9190 $\pm$ 0.0131
Exp2 (GeHirNet)	0.9647 $\pm$ 0.0054	0.9644 $\pm$ 0.0058	0.9294 $\pm$ 0.0114
Exp3.1 (GeHirNet*)	0.9646 $\pm$ 0.0023	0.9641 $\pm$ 0.0024	0.9289 $\pm$ 0.0047
Exp3.2 (GeHirNet**)	0.9724 $\pm$ 0.0037	0.9723 $\pm$ 0.0038	0.9449 $\pm$ 0.0074

TABLE III
ABLATION STUDIES

Ablation	Accuracy	F1	MCC
Sex as input feature	0.9556	0.9551	0.9109
Increase network depth	0.9600	0.9596	0.9201

TABLE IV
COMPARISON OF DISEASE-SPECIFIC ACCURACY BETWEEN OUR MODELS
AND PRIOR SOTA

Disease	Prior	Baseline	GeHirNet	GeHirNet**
Dysphonia	0.99 [32]	0.986	0.990	0.992
Laryngitis	0.983 [33]	0.997	0.997	0.998
Vocal Cord Paresis	0.941 [34]	0.989	0.994	0.996
COVID-19	0.999 [35]	0.979	0.986	0.991
Parkinson's	0.997 [36]	0.995	0.997	0.997
ALS	0.997 [37]	1	1	1

clusters for several diseases, indicating that sex-related acoustic differences are reflected at the individual level and motivate the need for a sex-aware modeling approach. Table V shows that ALS and healthy controls exhibit significant differences, with females having higher spectrogram power than males ($\Delta = +4.92$ dB for healthy, 95% CI [4.509, 5.335]; $\Delta = +4.39$ [2.295, 6.587] dB for ALS). In contrast, COVID-19 and Vocal Cord Paresis showed smaller differences ($\Delta = -1.52$ [-2.218, -0.845] dB for COVID, $\Delta = -0.92$ [-1.676, -0.162] for Vocal Cord Paresis, while Parkinson's, Dysphonia, and Laryngitis exhibited minimal sex-related variation. These observations suggest that some sex-related acoustic differences are disease-specific and motivate the need for a sex-aware modeling approach.

To investigate how the model captures such differences, we analyzed sex-based feature representations using CKA similarity scores (Fig. 3). Shallow layers exhibited high similarity across sexes, indicating that early layers extract universal vowel characteristics shared by males and females. Deeper layers, however, showed lower similarity, suggesting that later layers capture more sex-specific, pathology-relevant patterns. Specifically, shallow layers encode general spectro-temporal features such as spectral envelopes [38] and formant ratios [39], which are relatively stable across sexes. In contrast, deeper layers capture clinically meaningful differences: males typically exhibit lower fundamental frequency (F0) and formant values due to longer and heavier vocal folds [40], whereas females show higher F0 and distinct glottal dynamics. These patterns align with established physiological differences

in voice production and provide a mechanistic explanation for the benefits of a sex-aware hierarchical design.

Finally, we evaluated fairness across sexes using EOdds and Macro-TPR disparity. EOdds differences between models were not statistically significant (Friedman test, $p > 0.05$). For Macro-TPR disparity, overall values decreased from 0.067 in the baseline model to 0.031, 0.048, and 0.037 in the **GeHirNet**, **GeHirNet***, and **GeHirNet**** variants, respectively, indicating relatively balanced performance across sexes. These results suggest that explicitly modeling sex produces comparable fairness.

Taken together, these results demonstrate that incorporating a sex-aware stage not only improves overall classification performance but also addresses bias in detecting pathology, even when low-level acoustic differences are subtle. The observed sex differences in Mel spectrograms and the corresponding feature representations validate the rationale behind GeHirNet and explain why it outperforms sex-agnostic systems in achieving more precise and equitable pathology detection.

F. Related Work

Prior research on hierarchical voice pathology classification, such as Cordeiro's three-class system achieving 83% accuracy [41], has not accounted for sex differences. Our work advances this paradigm by introducing sex-based hierarchical classifiers that explicitly address class imbalance—a critical issue exacerbated by sex disparities.

Unlike previous hierarchical frameworks that used disconnected modular blocks (e.g., separating sex detection from pathology classification) [11], our two-layer model integrated sex representations directly into multi-class pathology detection. We implemented an end-to-end framework where the first layer outputs four sex-health classes, enabling disease classification in the second layer based on sex-aware representations.

Rather than treating sex and pathology as separate, sequential tasks, we explicitly embedded sex into the learning process, enabling the model to capture distinct, sex-dependent pathological features. By integrating sex directly into the classification pipeline, our approach mitigates performance disparities caused by sex-based variation.

IV. CONCLUSION

We acknowledge several limitations. Our study focuses on vowel /a/ recordings, which may limit generalization to more complex speech. The use of multiple datasets introduces variability in recording conditions, making it challenging to fully disentangle disease-specific features from environmental artifacts. While the hierarchical framework is conceptually

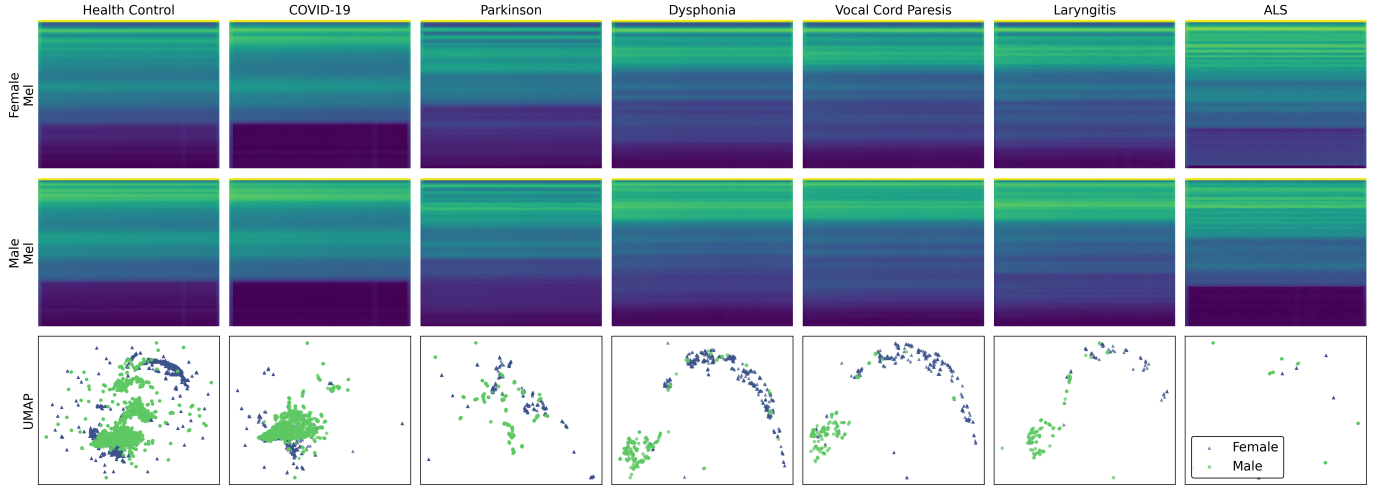


Fig. 2. Sex-specific acoustic characteristics across diseases. Rows 1–2: Average Mel spectrograms for female (top) and male (middle) participants, computed per disease. Row 3: UMAP embeddings of individual samples for the same diseases, with markers indicating sex.

TABLE V
SEX DIFFERENCES IN MEL SPECTROGRAM POWER ACROSS DISEASES.

Disease	Female (dB)	Male (dB)	Δ (F-M, 95% CI)
Health	-9.63 ± 9.49	-14.55 ± 9.16	$+4.921[4.509, 5.335]^*$
COVID-19	-22.38 ± 7.22	-20.86 ± 7.50	$-1.523[-2.218, -0.845]^*$
Parkinson's	-2.80 ± 3.74	-2.99 ± 3.14	$+0.192[-0.218, 0.602]$
Dysphonia	-2.69 ± 2.49	-2.96 ± 2.33	$+0.267[-0.139, 0.674]$
Vocal Cord Paresis	-2.56 ± 3.42	-1.64 ± 4.02	$-0.919[-1.676, -0.162]^*$
Laryngitis	-3.05 ± 3.04	-2.43 ± 3.42	$-0.618[-1.623, 0.387]$
ALS	-5.23 ± 7.63	-9.62 ± 4.55	$+4.393[2.295, 6.587]^*$

* indicate statistically significant differences ($p < 0.05$).

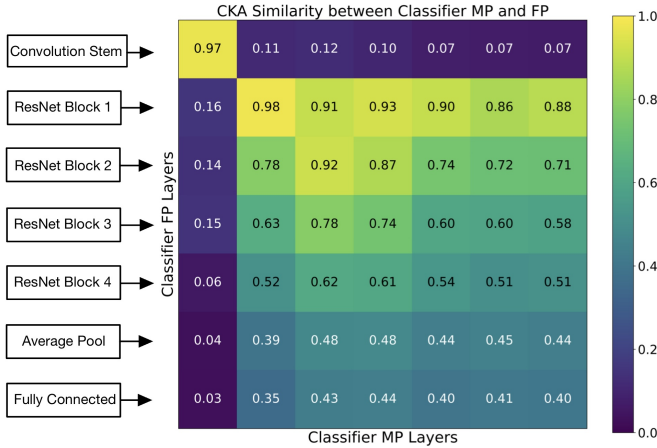


Fig. 3. CKA scores between Classifier FP and MP across ResNet layers: convolution stem, four residual blocks, pooling, and fully connected.

architecture-agnostic, we evaluated it with a single backbone (ResNet-50), and extending it to other disease classifiers is an important future direction. Finally, the clinical interpretation of learned frequency-region patterns remains limited, leaving mapping to pathophysiological mechanisms for future work.

Despite these limitations, our work makes significant contributions by addressing class imbalance and incorporating sex-

dependent features in voice data. Our two-stage hierarchical architecture first separates healthy and pathological samples in a sex-aware manner, then performs disease-specific classification. This design overcomes critical limitations of traditional single-stage approaches, which often struggle with rare-disease underrepresentation and ignore sex-based vocal variations.

Through controlled experiments, we demonstrate that the two-stage architecture alone outperforms prior methods, even without augmentation. Additional gains are achieved with targeted techniques such as time-warping, which improve accuracy for underrepresented classes. By systematically addressing both class imbalance and sex-based variability, our framework enhances diagnostic accuracy, reliability, and fairness in data-limited scenarios. These results highlight the potential of scalable, cost-effective voice pathology screening and underscore the importance of sex-aware modeling to mitigate underdiagnosis or misclassification in clinical deployment.

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